

Design Alpha Release

Due: In Studio, week of 2025-11-03 **Weight:** 10% **Submitter:** Team

Assignment Overview

An “Alpha release” is an early review of design concepts. It allows teams to get early feedback on their initial design concepts and their design process so far. In some engineering workplaces, people have what is called a “bench review”—where a supervisor or small group will discuss progress on a project gathered in a board room, or quite literally at a work “bench.” Preparation for such a review is minimal – it could literally be called on the spur of the moment (like a pop quiz). If you have been working on your project, you should not find the experience difficult; in fact, if you and your team have been actively participating in studio you may not need any additional preparation at all!

Our design Alpha release focuses on the results of teams’ diverging work in an informal, dynamic, semi-structured conversation. Alpha will consist of a 12-minute conversation between your team and members of the teaching team. As a team, your focus should be on knowing what you have done, how you have done it, and what your next steps need to be. The conversation will focus on understanding and critiquing your design process, and the design concepts you have developed so far. During the conversation the teaching team will direct questions at specific team members.

Assignment Stakeholders

There are four main stakeholders (or groups) for this assignment:

- Your team, who must demonstrate having engaged in credible engineering design activities and having developed a range of design concepts worthy of testing.
- You, individually, as an aspiring engineering designer who must develop both individual and team engineering design skills, and who needs to begin to differentiate themselves from their peers through their unique perspectives.
- Your teaching team, who is responsible for helping you to refine and respond to your opportunity with credible engineering design, and for guiding you.
- The stakeholders from your design brief, who by definition have an interest in both the eventual design concept and the process of its development.

Structure of Alpha Release

At this stage, you are expected to show four main aspects of your design work:

1. Design brief interpretation: you have to decide how to approach your chosen design brief as a design team. Whether you wrote the brief yourselves or are choosing another one, you may need to re-scope or reframe, and you will certainly need to refine the original opportunity. This likely will involve **research** to enable you to:
 - revise the stakeholders, defining them more precisely,
 - narrow the scope to a subset of the earlier scope,
 - **focus and refine the objectives into a set of requirements and evaluation criteria.** This may require adding/dropping a goal/objective, making objectives more precise, etc.
 - redefine metrics so that you can actually test your possible design alternatives (once you have them).

2. Process: in studio during the week of 2025-10-13, you will be guided through an initial cycle of idea generation and divergence. This will likely be done rapidly and without research. By Alpha, you should be able to show **multiple iterations of diverging with research** in order to demonstrate that you have approached the opportunity from multiple angles.

You should also be prepared to demonstrate your team's strategy for working and decision making. There is no "right" answer here, but you should be able to show and explain how your team develops ideas, makes decisions, and ensures equitable contributions from all members.

3. Design concepts: at this stage, we expect you to have at least **$n+1$ concepts** that you are considering, where n is the number of members on your team. Good concepts are those that are legitimately worth your consideration, and that appear, at least for now, to meet your requirements. At this stage, the representations of many of these concepts can be very sketchy (literally, they can be sketches).
4. Next steps: at this stage, you should be able to tell the teaching team what your **biggest questions** about each of your design concepts are (but NOT the answers to these questions!). You might be able to identify which metric(s) will be critical (i.e., which ones will really tell whether your idea will work). You might even be able to say how you could "proxy" a test for these metrics given your available time and equipment.

The teaching team will ask questions (to the team or specific individuals) related to the four aspects above. Do not plan to deliver a presentation on any of these aspects. The Alpha Release is Q+A only. Discussing these four aspects of your design work should allow each person to share in the conversation and demonstrate their understanding of the team's work.

You do not need to bring any specific visual aids beyond some prototype(s) for each of your design concepts, though you should have evidence ready to back up claims about your design process. This could be anything from meeting minutes to photographs of team activities, or any other evidence you have collected about your design process so far this semester. You should not be creating new "evidence" specifically for Alpha! Instead, consider spending time to organize your existing evidence for easy access during the Q&A. As a constraint (discussed below), you must bring at least one low-fidelity physical prototype to Alpha, and you should be ready to defend your choice of approach to prototyping your different concepts, as well as to share some preliminary insights you have gained so far through your various prototypes.

Your team will need to work collaboratively on your project so that knowledge of different aspects is shared and each team member is able to discuss **all** aspects of the design work. It is not acceptable for each team member to have "their concept" or "their prototype" that they work on in isolation. Although it can be reasonable to have **some** of the prototyping work be done individually, all team members must have understanding of and involvement in the development of **all** concepts.

Assignment Expectations

Requirements language in this and following sections are to be interpreted as described in RFC 2119.

Objectives

1. Thoughtful interpretation of the design brief
 - Explain and justify major deviations from the design brief, and how your team is approaching the opportunity as represented by a set of requirements and evaluation criteria.
2. Intentional process in divergence
 - Demonstrate individual and shared knowledge of the design process across its aspects.
 - Show insights gained through initial prototyping such as sketches, and your one simple physical prototype.
 - Demonstrate the use of, and learning from, formal engineering tools in support of your design activities to allow divergence.
3. Quality and diversity of concepts
 - Show $n+1$ credible¹ design concepts that address the opportunity, where n is the number of members on your team.
 - Represent and communicate design concepts such that audiences can recognize and understand the key features of each design concept, and the design decision(s) associated with those features.
- 4. Quality of Communication
 - Utility of team and individual contribution to the discussion.
 - Demonstrate effective teamwork that shows how individual skills and abilities were integrated to develop your team's results.
 - Communicate clearly using a collection of relatively unstructured oral, visual, physical, and written representations.

Requirements

For your Alpha Release, your team:

1. **Shall** demonstrate that you have diverged such that you have at least $n+1$ credible design concepts for further development and consideration.
2. **Shall** demonstrate that your process has made use of divergence tools (e.g. SCAMPER, functional decomposition, Brainwriting 6-3-5, etc.) to overcome cognitive biases.
3. **Shall** have prototypes for all design concepts being discussed, including **at least one** (1) **low-fidelity physical** prototype.

¹ A “credible” design concept at this stage is one that you anticipate is verifiable

Evaluation Criteria

The evaluation criteria reflect the goals of the overall conversation. No individual is responsible for all of these, but each is responsible for ensuring these characteristics are reflected in the overall discussion.

For all of the characteristics, the criteria prefer “more,” “higher,” or “greater”:

1. Quality of engineering argument, with emphasis on justifying claims through credible use of engineering-appropriate sources.
2. Usefulness of the engineering requirements and evaluation criteria to support decision making about design concepts.
3. Quality of the learning generated by applying engineering tools for diverging and representing (e.g. prototyping) candidate design concepts.
4. Quality of the concepts under consideration, as demonstrated by the representations, preliminary assessment against requirements and evaluation criteria, diversity of approaches, and plans for development.
5. Effectiveness of teamwork during the discussion that demonstrates both integrated and mutually supportive contributions.
6. Clarity and integration of oral, visual, physical, and written communication.
7. Ability to participate in meaningful, professional-style conversation with key stakeholders.